Challenge in Enabling Pipeline Repurpose for CO2 Transportation

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1 ABSTRACT

As the global energy sector advances Carbon Capture, Transportation, and Storage (CCT&S) initiatives, repurposing existing gas pipelines for CO_2 transport offers a cost-effective and sustainable solution. This study presents a comprehensive feasibility and qualification assessment for converting a 30" x 220 km offshore gas export pipeline in Malaysia to transport dense-phase CO_2 from CCS hubs in Peninsular Malaysia to offshore depleted reservoirs.

The evaluation spans four key technical areas:

1. **Fracture Toughness Qualification**
   Due to missing manufacturing certificates, a statistical analysis of 5,000 data points from similar pipes was used to infer Charpy V-Notch (CVN) values. Results showed higher-than-standard toughness, except for a 5 km section (KP4–KP9), which requires replacement due to lower resistance under specific conditions.
2. **Corrosion Assessment**
   Using OLGA simulations, corrosion rates were predicted based on CO_2 composition and operating conditions. No water dropout was observed under normal or blowdown scenarios, indicating minimal corrosion risk.
3. **Depressurization Impact Analysis**
   The study assessed rapid depressurization effects on temperature, phase behavior, and solid formation risks. Simulations confirmed that a 6–7-day blowdown is feasible without breaching the pipeline's minimum design temperature of 0°C.
4. **Dropped Object**
   A risk assessment using DNV-RP-F107 methodology evaluated threats from dropped and dragged anchors. Results showed that the risk of failure remains within acceptable industry limits, supporting the pipeline's continued use.

As findings demonstrate, with appropriate material qualification, corrosion management, operational controls, and external risk mitigation, re-purposing the existing export gas pipeline is technically a viable option for dense phase CO_2 transportation. This work provides a robust technical framework and practical insights for similar repurposing projects, advancing the deployment of CCS infrastructure in Southeast Asia and beyond.

2 INTRODUCTION

PETRONAS has set its Net Zero Carbon Emission (NZCE) by 2050 with medium term target to limit its carbon emission to 49.5MTPA by 2030. Carbon Capture & Sequestration is one of the identified decarbonisation pathways. PETRONAS, together with an international partner signed an MOU to study the development of a legacy gas field and store CO2 in its depleted reservoir. Apart from reservoir, gas processing platform and 230km x 30in gas pipeline are to be repurposed.

Design Parameters	Value
Commission Year	1996
Current Service	Dry Gas
Material	Carbon Steel
Material Grade & Specification	X65 API Spec 5L
SMYS (MPa)	448
UTS (MPa)	530
Manufacturing Process	LSAW (Cold Expanded)
Wall Thickness Z1 and Z2 (mm)	20.62, 28.58
Corrosion Allowance (mm)	None
Water Depth (m)	0-70

Upon completion of conceptual study by PETRONAS and its partner, a project risk assessment, was conducted to identify remaining gaps which translated to this work.

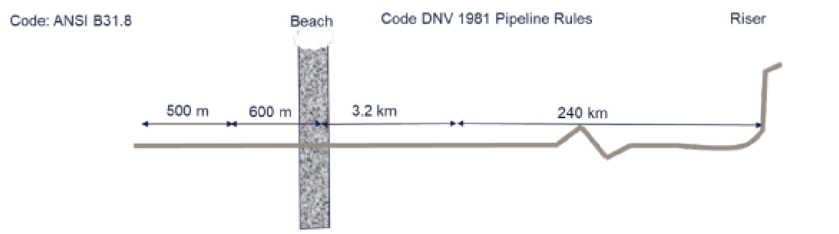
Among the challenges addressed in the risk assessment include:

- Unavailability of mill certificates that contain key data i.e. CVN to assess running ductile fracture (RDF) propagation.

  All efforts have been made to locate remaining mill certificates for zone 1 section (Sector IV) of the pipeline but to no avail. Hence, the key toughness parameters, CVN, for the majority pipeline section is unknown. Credible technical assumptions by material experts are needed to address this.

- Potential presence of longitudinal crack following technical assessment performed in 1996 when it was found during the installation.

  Upon finding the longitudinal crack during the installation, DNV was hired to perform engineering criticality assessment (ECA) to ensure that the crack would not propagate throughout 30 years of pipeline operational loading. DNV had proven that pipeline was safe. However, the pipeline operator had taken a conservative conclusion; "it must be assumed that the probability exists for all pipes to contain similar crack, particularly the doubly expanded pipe."



Sector	I	II	III	IV
Wall thickness (mm)	28	28	28	20
Min temp. (°C)	-20	20	20	20
Hydrotest pressure (MPa)	22.95	22.95		18.796
Operating pressure (MPa)	13.792	13.792		13.792
CTOD -20°C /+ 20°C (mm)	0.06/0.11	0.11	0.11	0.44

Figure 1 Schematic of 30" x 230km Gas Pipeline

3 TECHNICAL EVALUATION

An expert study had been initiated to address related concerns on

I. fracture toughness value, CVN of zone 1 section where the mill certificates are unavailable
II. studying the growth of longitudinal crack, if any, to reach critical limit and initiates running ductile fracture (RDF).

As for other technical concerns, the appointed expert studied

III. corrosion rate estimation of the pipeline CO2 specification
IV. pipeline depressurization impact to pipeline
V. possibility of dropped and dragged that would initiate RDF from external.

3.1 FRACTURE TOUGHNESS

SUPERB JIP report that contains database of 5000 linepipes information covering sizes from 8” to 56” and thickness range from 13.0mm to 36.8mm have been referred to. These linepipes include UOE, LSAW and seamless that being manufactured in 1990s. A comparison has been made against toughness property of the gas pipeline which led to an observation that its CVN values are on the upper shelf range. The mean CVN value at the material transition temperature, -29°C for the pipeline is 175J compared to database, 130J. Hence, the gas pipeline should be expected to demonstrate higher CVN value during operation; the lowest analyzed temperature for controlled blowdown is 15°C.

However, actual representative test is highly recommended to prove the claim. Adding further, the expert opined that 20.62mmWT section would have better toughness compared to 28.58mmWT due to finer microstructure from the plate rolling process.

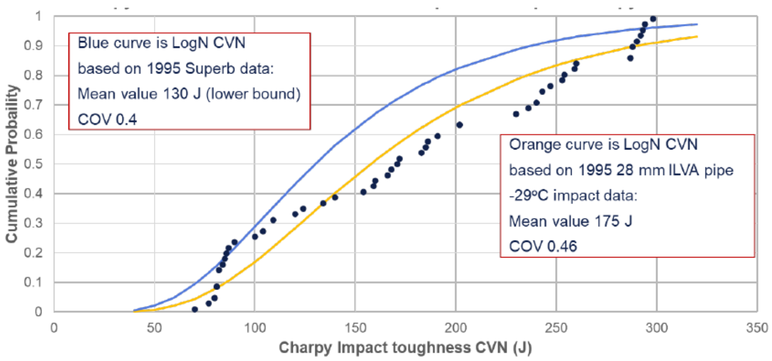


Figure 2 Toughness value comparison with SUPERB JIP datapoints

Minimum required toughness, CVN, to arrest RDF propagation had been assessed as per DNV-RP-F104. Based on input CO2 pipeline specification, at the lowest saturation pressure, P_{sat} 70barg, the minimum required CVN is 320J. However, this case only represents a single point; pipeline inlet. PETRONAS had extended the analysis by calculating minimum required CVN value along pipeline length. Every analyzed case is input with associated temperature at concerned pipeline KP to obtain P_{sat}. Conservatism has been included where similar pressure of 120barg has been used regardless of temperature. Further, based on DNV-RP-F104, minimum required CVN value is determined when it intercepts with the boundary of ‘Evaluation based on small scale testing’.

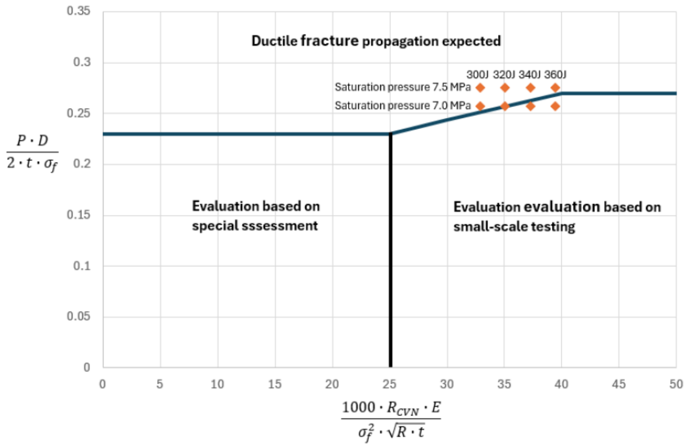


Figure 3 Toughness value evaluation as per DNV-RP-F104

Based on internal technical analysis, it requires as high as 460J (max) to arrest RDF propagation given lower wall thickness, 20.62mm compared to onshore section wall thickness, 28.58mm. As temperature lower, lower P_{sat} is observed which reduces the toughness required to arrest RDF. Despite there are 2 ways to obtain minimum toughness, CVN value required to arrest RDF propagation, estimated CVN value provided by DNV methodology has been adopted. The technical criteria and assumptions are based on observations from large scale testing.

Section	Onshore		Nearshore	Offshore (20.62mmWT)					
Estimate KP	0	2	3.2	4-5	6-7	8-9	10-11	12-29	30-200
Temperature	60	55	50	45	40	35	30	25	20
E - BTCM									
eBTCM Saturation Pressure , barg	75.64	77.5	77.49	75.21	71.45	66.71	61.55	56.35	51.3
eBTCM CVN Value , J	129.8	147.2	147.1	126.3	101.9	80.1	62.7	49.2	38.8
DNV F104									
DNV Saturation Pressure, barg	290 J (mill cert zone 2)			75	75	70	64	64	64
DNV CVN Value, J				>290			116	116	116

Figure 4 Minimum required CVN value to arrest RDF propagation.

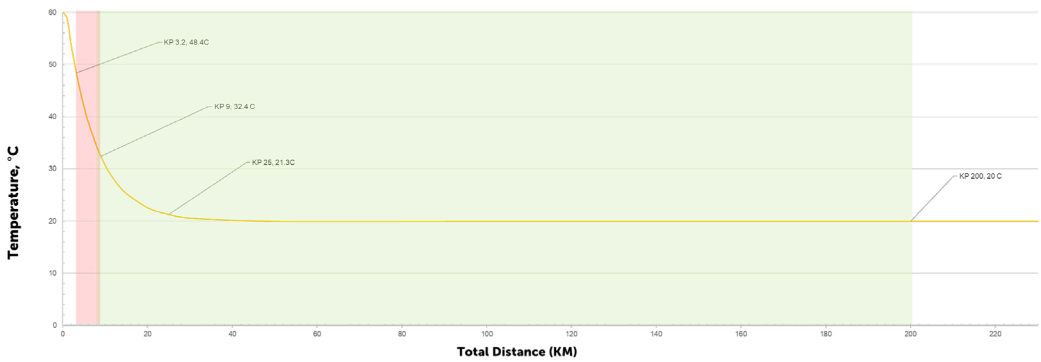


Figure 5 Indicative sectional replacement; KP4-KP10.

3.2 RDF Initiation

Longitudinal crack can be the initiator RDF. Hence, for every crack, it needs to ensure that it will not grow critical and unstable.

Particularly for this pipeline, during installation in 1996, a longitudinal crack (8.8mm deep, 100mm length) was found at HAZ, externally near pipe end, in 1 of 4, 28.58mmWT linepipethat had been cold expanded twice. The crack was later grounded and laid to the seabed.

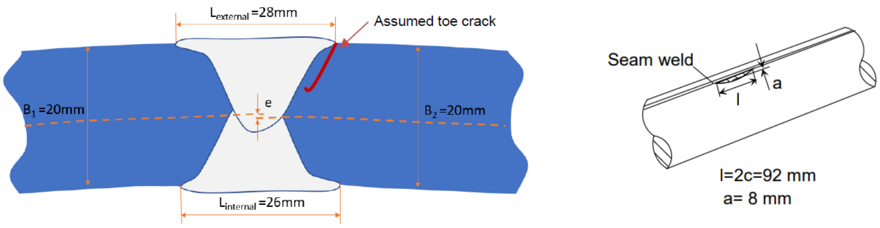


Figure 6 Longitudinal Crack Schematic

The most likely root cause was delayed hydrogen cracking. Hence, it had been concluded, “it must be assumed that the probability exists for all pipes to contain similar crack, particularly doubly expanded pipe” as taken from the investigation report.

The potential existence of longitudinal crack in the pipeline had been discussed at length if it would grow critical during the operation of dense phase CO2. Hence, expert had proposed to run fatigue analysis using available operation data of the pipeline in current usage, gas transportation. The pressure data is hovering around 80 to 100barg. It may be argued the relevancy of the data to its future usage. However, it is the best realistic data available to provide good understanding of the pipeline resistance towards fatigue.

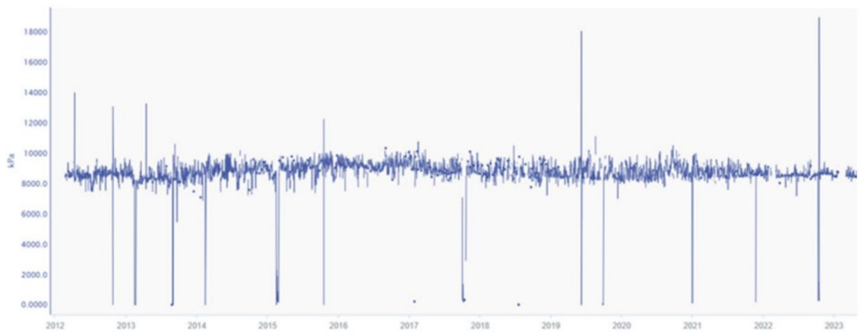


Figure 7 Pressure Record of the Pipeline from 2012 to 2024

Hence, it was found, in case of similar crack exist in any of linepipe, the crack will not grow critical, even when exposed to 300 years of operational loadings.

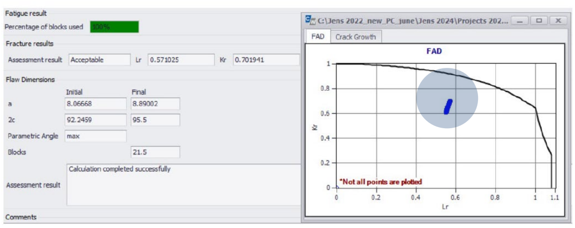


Figure 8 ECA Assessment based on fatigue loading of 10x pipeline design life (300years).

3.3 Corrosion Assessment

Theoretically, without the presence of free water, dense phase CO2 is not corrosive to carbon steel pipeline. However, the CO2 composition may contain traces of impurities that can significantly impact corrosion behaviour especially water precipitation. Therefore, the effect of water content and traces of MEG and SO2/NO2 were studied. These selected impurities are likely to exist in the pipeline either during commissioning or during abnormal operation. The corrosion assessment established corrosion rate given operating data and fluid composition in CO_2 services with the effect of water content, MEG and SO_2/NO_2 gases.

Table 1 Target CO2 Specification for Pipeline Transport

Component	Dense Phase
CO2	> 96.0%
H2O	< 200ppm < 630ppm if no NOx and/or SOX present
H2	<1.0%
Combine volatilities: H2,C1,N2 & Air	<4.0%
CO	< 1000ppm
O2	< 10ppm
Combines H2S, COS, CS2 & Mercaptans	< 20ppm, partial pressure not to exceed 0.05 psia
C2-C5 Hydrocarbons (including Olefins)	< 0.5% (5000ppm)
Nox	< 10ppm
Sox	< 10ppm
Ammonia	< 10ppm
Combines BTEX	< 100ppm
Hydrogen Cyanide	< 1ppm
Amines	< 10ppm
Mercury	< 0.01 µg/Nm3
Methanol	< 500 ppm
MEG/TEG	0.3 gal/MMSCF

Table 2 describes various operating conditions of CO_2 pipeline including normal and abnormal. Target specification outlines water content <200ppm. OLI software estimated the minimum water content for free water to dropout hence led to corrosion. The corrosion rate for each case varies, depending on presence of MEG. In general MEG reduces water saturation point.

The lowest content for free water to dropout is during abnormal condition 560ppm (without MEG) and 100ppm (with MEG). Regardless the lowest corrosion rates are observed in during blowdown in gas phase 6.68mm/year (without MEG) and 5.55mm/year (with MEG), the blowdown itself is a transient process. Hence, abnormal operation in general would require corrosion inhibitor.

During normal operation, it takes 12-13 times breach of water limit for free water to dropout. Once breached, the estimated corrosion rates are 50.07mm/year (with MEG) and 48.40mm/yr (without MEG) respectively. The addition of MEG in the system slightly reduces the corrosion rate for all operating conditions, however, its effect on reducing the water saturation limit is still concerning.

Hence, keeping the water content within the target specification will be critical to protect the pipeline. Once corrosion take place, it will be severe and affect the pipeline integrity.

Lastly, effect of SOx and NOx were simulated based on maximum target limit at <10ppm each. The result of the simulation (without MEG) showed that the SOx/NOx will require very small amount to react with free water for acid to drop-out causing pH of the dropout below 0 and thus resulting in extreme corrosion rates which are unable to be determined using OLI modelling.

3.4 Depressurization Impact Analysis

The depressurization assessment is conducted to study the operating parameters under full blowdown. This includes investigating the potential impact on minimum design temperature, the occurrence of phase changes, the risk of internal corrosion due to water dropout, the possibility of solid formation and adequacy of the event system to safely manage pressure relief. These factors are critical to ensuring the pipeline can withstand transient operation conditions without compromising structural integrity or safety.

Full depressurization of any CO2 pipeline should be considered as a rare event; hence CO2 pipelines should be operated to minimize the number of planned depressurizations.

Minimum Temperature

The transient simulations calculated that a blowdown of the pipeline over approximately 6-7 days is feasible without conflict with a minimum pipeline design temperature of 0°C.

With regards to pipeline temperature response associated with the blowdown operation, the pipeline benefits from higher ambient seabed that supress further temperature reduction, to below 0°C. The temperature reduction associated with phase transition during the blowdown operation is effectively compensated by heat ingress from ambient. This benefit is closely linked to the blowdown rate, implicitly the time of the blowdown operation.

For dense phase CO_2, the depressurization operation is split into the following three phases, a) initial liquid expansion, b) boiling (two-phase), and c) gas expansion phase. Figure 9 shows pressure response reflecting schematically the three phases of the operation.

Table 2 Summary of Corrosion Estimation Assessment

No	Operating Condition	Pressure (Bara)	Temperature (°C)	CO2 Phase	Specific water content (mol%)	Without MEG, Without SOx/NOx		With MEG (0.3gal/mmscf), Without SOx/NOx	
						Critical water content for water dropout (mol%)	Corrosion rate @ critical water dropout	Critical water content for water dropout(mol%)	Corrosion rate @ critical water dropout
1	Normal (Pipeline inlet)	107	50	Supercritical	0.0199329	0.267	50.07	0.249	48.40
2	Normal (Middle pipeline)	109	25	Liquid	0.0199329	0.128	12.60	0.125	12.60
3	Abnormal (Start of blowdown)	112	20	Liquid	0.0199329	0.108	8.77	0.106	8.72
4	Abnormal (Blowdown-Dual phase)	75	25	91% mol liquid/9%mol gas	0.0199329	0.115	12.54	0.110	12.28
5	Abnormal (Blowdown-Dual phase)	70	25	84% mol gas/16%mol liquid	0.0199329	0.087	12.54	0.079	11.99
6	Abnormal (Blowdown-Gas phase)	55	16.5	Gas	0.0199329	0.056	6.68	0.010	5.53

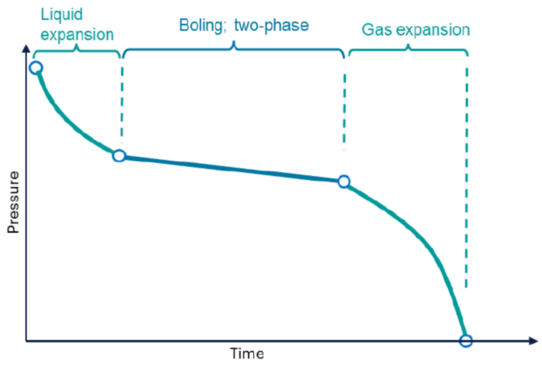


Figure 9 Phases of Blow-down Operation

The pressure response for the first 1-2 days of the operation coincides with the expected pressure response for liquid expansion. For this period the pipeline is expected to remain in liquid filled, hence the volume of product removed is compensated by the compressibility of the liquid phase.

The relatively flat plateau at the vent location of approximately 70 bar i.e. P_{sat} reflects the two-phase boiling stage that occurs close to "isothermal", where the heat of evaporation associated with phase transfer from liquid to gas phase is compensated by heat ingress from ambient to the pipeline. This period is approximately 2-5 days. In this period, it is foreseen that most of the product will boil off to the gas phase.

The final stage lasting from day 5 to day 7, is foreseen to be dominated by gas expansion, implicitly decompression of the CO2 gas phase.

The above reflections should be taken only as a high-level assessment which more accurate case of blowdown to be determined in the later stage of the study.

Water Saturation

For the pipeline depressurization is conservative to assume that the pipeline inventory is based on target conditions for water specification for dense phase, i.e. 200 ppm. The composition is specified as 96% CO2 with a maximum of 4% non-condensable components (Argon, Oxygen, Hydrogen, Nitrogen, Methane)

The water saturation is a function of pressure, temperature and product composition. During the pipeline depressurization, the pressure will gradually reduce. The operation is also normally associated with a temperature reduction. Specific to the pipeline operation, the effective heat ingress compensates to suppress the temperature reduction. Based on table 2, there is a potential of water dropout during gas expansion i.e. blowdown gas phase with 0.3gal/mmscf MEG where water saturation limit is predicted lower than the target water limit, 100ppm and 200ppm respectively.

Therefore, the risk of hydrate formation during the blowdown operation will be further investigated in the next stage of study.

3.5 Dropped Object

The primary objective is to understand the frequency of dropped objects such as anchors or equipment striking the pipeline and initiating mechanical damage. These impacts can result in localized deformation or cracking, which under dense-phase CO_2 conditions may lead to crack propagation in the form of running ductile fracture (RDF). To address this, a frequency analysis was conducted as per DNV-RP-F107 to quantify the likelihood of such events and assess their consequences.

The vessel traffic data ocean vessels passing through the vicinity of pipeline use in the assessment for Year 2022, which consists of a total of 2,640 vessels per year, had been applied. An average speed of 8 knots is assumed for vessels traversing across the pipeline.

There are three Single Anchor Leg Mooring (SALM) in the vicinity of pipeline route as per Figure 10. Due to the extent of the anchor drop zone radius, only the anchor dropped zones at refinery and SALM1 may expose pipeline to a potential hit, thus the evaluation would only assess vessels at refinery and SALM1.

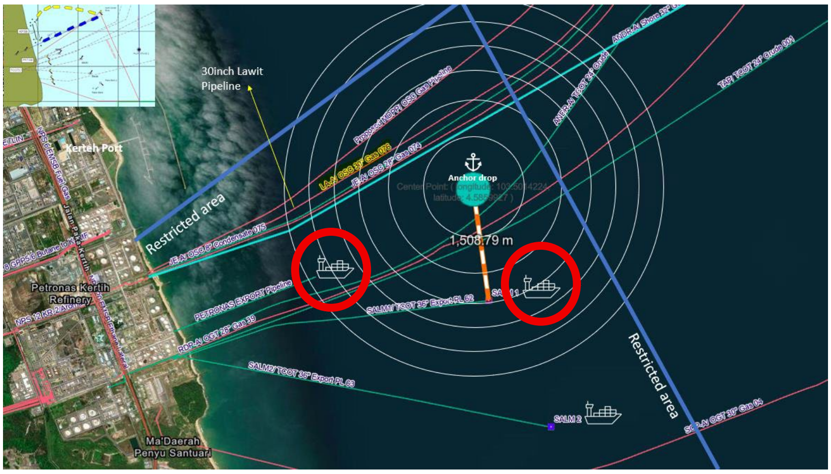


Figure 10 Potential Anchor Drop Zone

The impact energy is calculated based on parameters of the pipeline. The impact energy required to produce each level of relative dents is tabulated in Table 3. The result shows the pipeline will be with impact of 253MJ.

Table 3 Energy Required to Damage The Pipeline

Relative Dent (%)	Impact Energy (MJ)	Damage Description
<5	<32	Minor damage
5 – 10	32 - 89	Major damage. Leakage anticipated
10 -15	89 - 164	Major damage. Leakage and rapture anticipated
15 – 20	164 - 253	Major damage. Leakage and rapture anticipated
>20	>253	Rupture

Based on input data above, the results of the calculation of the frequencies of interaction for dropped anchors are tabulated in Table 4.

Table 4 Dropped Anchor Calculation

Parameter	Bulk Carrier	Tanker	Container	Tanker at SALM	Total/year
No of ship/yr	800	1040	800	227	
Average Mass of Anchor(kg)	8800	16,200	8,055	14,700	
Terminal velocity (m/s)	9.08	12.58	8.66	11.95	
Impact Energy (MJ)	399	1,410	332	1,188	
Relative dent (%)	>20 (Rupture)	>20 (Rupture)	>20 (Rupture)	>20 (Rupture)	
Critical distance (exposed pipeline to anchor) (m)	6.5	7.93	6.5	7.67	
Frequency of dropped anchor (/yr)	4.45E-07	7.06E-07	4.45E-07	1.32E-07	1.73E-06

The impact energy is driven by the terminal velocity of the anchor which is influenced by the size of the anchor. Therefore, the tankers imposed the highest impact energy. It can be concluded that all dropped anchors results in a >20% relative dent in the pipeline which means a rupture of the pipeline.

The overall frequency of dropped anchors is approximately 1.73E-06/year. Based on the frequency ranking table per Section 5.8 of DNV-RP-F107, this is defined as “Likelihood of event considered negligible”.

The frequencies of interaction for dragged anchors are calculated and tabulated in Table 5.

Table 5 Dragged Anchor Calculation

Parameter	Bulk Carrier	Tanker	Container	Tanker at SALM	Total/year
No of ship/yr	800	1040	800	227	
Average Mass of Anchor(kg)	8800	16,200	8,055	14,700	
Terminal velocity (m/s)	2.06	2.06	2.06	2.31	
Impact Energy (MJ)	21	39	19	45	
Relative dent (%)	<5	5-10	<5	5-10	
Critical distance (exposed pipeline to anchor) (m)	70.53	85.32	68.62	82.77	
Hooking probability	100%	100%	100%	100%	
Frequency of dragged anchor (/yr)	9.65E-06	1.52E-05	9.39E-06	2.86E-06	3.71E-05

It can be concluded that dragged anchors results in a <5 or 5 to 10% relative dent in the pipeline which means a minor or major damage of the pipeline.

The overall frequency of dragged anchors is approximately 3.71E-05/year. Based on the frequency ranking table per Section 5.8 of DNV-RP-F107, this is defined as “Event rarely expected to occur”.

The results of the frequency analysis indicate that while the probability of rupture due to dropped anchors is negligible, dragged anchor events—though rare—could result in moderate damage. These findings support the implementation of targeted risk mitigation strategies to ensure long-term pipeline safety.

4 Conclusion

1. Upon benchmarking with the similar other similar linepipe data, it is observed that the pipeline has upper shelf toughness values, CVN.
2. The minimum required CVN value to arrest RDF propagation has been extended for whole pipeline length. Company suggested to sectionally replace KP 4-10 as the CVN value is lesser than the minimum required to arrest RDF propagation.
3. On RDF initiation, using the best available operating data, the pipeline can resist fatigue failure from longitudinal crack, at the safety factor of 10.
4. Specified water limit, <200ppm is well below saturation limit during normal and abnormal operations. It is wise to adhere to the water limit to avoid corrosion either due to the carbonic acid or acid drop out due to SOx and NOx.
5. The controlled blowdown is estimated for 7days. During this operation, the lowest temperature achieved is 15°C which above minimum design material temperature. The risk of hydrate appearing is to be further studied.
6. On frequency of dropped and dragged anchor, the frequency of both cases to happen has been assessed as 'rarely expected to occur'. However, the consequence of dropped and dragged anchor can be significant, ruptured and dented respectively. Historically, there was a case of dragged anchor in the area.
7. In conclusion, re-purposing the pipeline is feasible provided several activities to be carried out including material testing, in-line inspection to detect crack and ultimately sectional replacement.

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